ML MODEL DOCUMENTATION

- OVERVIEW:

The objective of this project was to develop a machine learning model that predicts whether a customer will make a payment during the current billing cycle.

, where:

- 0 = Customer did not make a payment
- 1 = Customer made a payment

- DATASET:

The dataset contains information for 1,000 customer accounts and originally consisted of 146 features. The variables include:

- Customer demographic information
- Account balances
- Payment history
- Delinquency information
- Contact history
- Credit bureau information
- Behavioural risk scores

After preprocessing and feature engineering, the final dataset contained 112 variables.

- DATA UNDERSTANDING:

Before building the ml model understanding data was important. The following checks were made:

- Dataset shape
- Data types

- Missing values
- Duplicate records
- Target variable distribution
- Numerical feature distributions
- Categorical feature distributions

This helped identify data issues before modelling.

- # EXPLORATORY DATA ANALYSIS (EDA):

EDA was performed for better understanding of customer behaviour and to identify useful patterns.

Target Variable Analysis:

The target variable used in this project is Current_Payment**.** The distribution of the target variable showed:

- No Payment = 741 customers (74.1%)
- Payment = 259 customers (25.9%)

This indicates that the dataset is moderately imbalanced.

Missing Value Analysis:

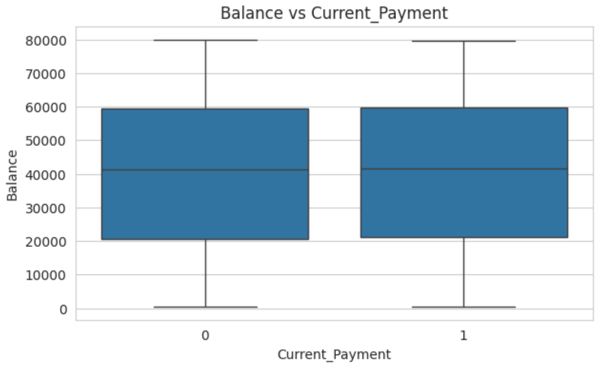
Each column was analysed for missing values. Variables containing more than 70% missing values were removed because they contained insufficient information for reliable modelling. The remaining missing values were handled during preprocessing.

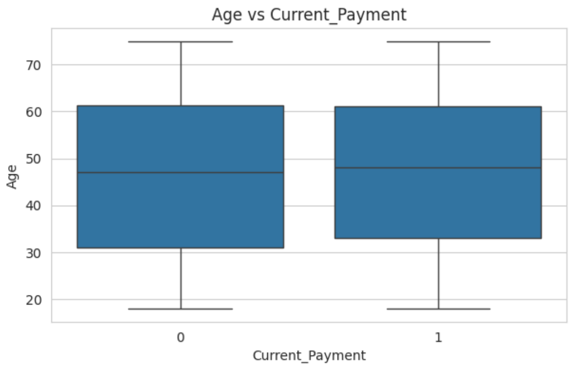
Duplicates:

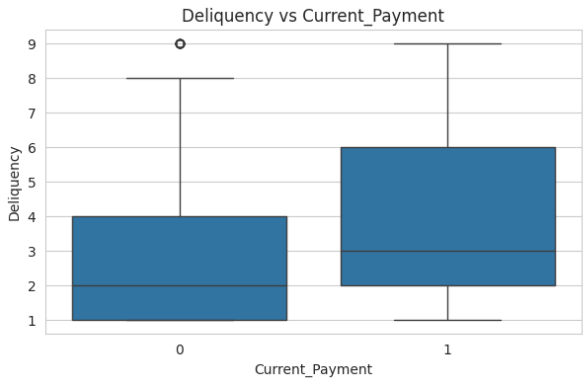
The dataset was checked for duplicate rows. No duplicate records were found.

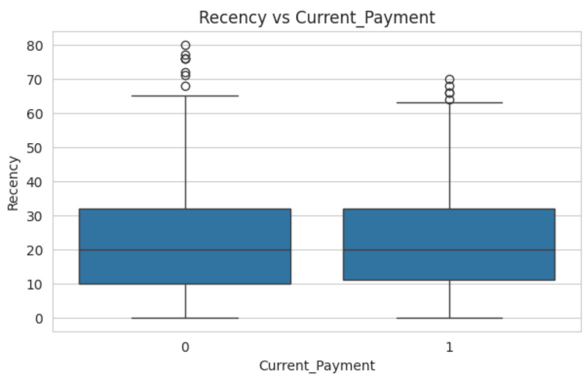
Numeric Column Analyis:

Numeric columns such as age, balance, recency, delinquency were plotted on the graph with the target variable for identifying patterns.



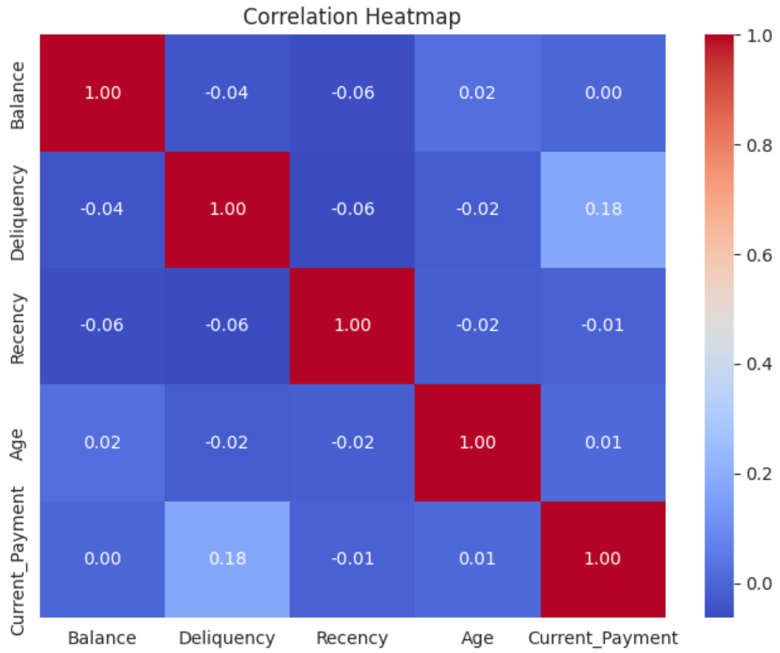






Correlation Analysis:

A correlation matrix was created to examine relationships between numerical variables. Highly correlated variables were carefully reviewed to avoid unnecessary redundancy.



Point Biserial Correlation:

Because the target variable is binary, Point-Biserial Correlation was calculated. This analysis helped identify which numerical variables had stronger relationships with customer payment behaviour. Variables such as Previous_Total_Due, Delinquency and Previous_Payment showed stronger correlations with the target.

- DATA PREPROCESSING:

Data Preprocessing was done because machine learning models perform best on clean and consistent data.

Removing High Missing Value Columns:

Columns containing more than 70% missing values like deceased information were removed because they were unlikely to contribute useful information.

Removing Data Leakage:

The following variables were removed:

- Current_Payment_Amount
- Current_Cure

Encoding Categorical Variables:

Categorical variables such as Gender and Marital Status were converted into numerical format.

Removing Near Constant Variables:

Variables containing almost identical values for every customer could not be useful and hence they were identified and removed.

- # TRAIN TEST SPLIT:

The cleaned dataset was divided into:

- Training Set (80%)
- Testing Set (20%)

The training set was used to teach the model. The testing set was kept separate. This ensures that model performance is evaluated using previously unseen data.

- # MODEL SELECTION:

A Random Forest Classifier was selected. Random Forest combines many decision trees to make a final prediction. Instead of relying on a single tree, multiple trees vote on the final outcome. This provides better accuracy.

- ## MODEL TRAINING:

The model was trained using the training dataset. During training, the Random Forest learned relationships between customer characteristics and historical payment behaviour. Once training was complete, the model generated probability scores that each customer would make a payment.

- ## MODEL EVALUATION:

The model achieved:

AUC: 0.639

PRECISION: 0.3302

RECALL: 0.6731

F1 SCORE: 0.4430

Confusion Matrix:

It showed:

1. True Positives: Correctly predicted payments
2. True Negatives: Correctly predicted non-payments
3. False Positives: Predicted payment but customer did not pay
4. False Negatives: Customer paid but model failed to identify them

- ## CROSS VALIDATION:

Mean AUC-ROC: 0.6466

Standard Deviation: 0.0248

This shows that the model performs consistently on different subsets of the data.

• FEATURE IMPORTANCE:

Random Forest provides feature importance scores, allowing us to identify which variables contributed most to the predictions.

The most influential features included:

- Account_Age
- Previous_Total_Due
- Case_Age
- PAttempts
- Never_Paid_Flag
- Contact_Score
- BehaviourRiskScore
- PaymentProjectionScore

• BUISNESS INTERPRETATION:

Instead of contacting every customer equally, businesses can prioritise customers with higher predicted payment probabilities, improving collection efficiency while reducing operational costs.

• CONCLUSION:

This project successfully developed a complete machine learning pipeline for predicting customer payment behaviour. The project included data cleaning, feature engineering, exploratory data analysis, model training, evaluation, cross-validation and feature importance analysis. The final Random Forest model achieved an AUC-ROC score of 0.6369 and demonstrated stable performance during cross-validation.